

Exploring Educational Potential of VR and Robotics: A Case Study from a Multinational Hackathon

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Abstract

This paper presents a reflective case study on the use of immersive technologies—virtual reality (VR) and robotics—within a cross-cultural educational hackathon. The 12-day program brought together interdisciplinary student teams from two countries to address a socially relevant challenge: how to promote physical activity among children on the autism spectrum. Anchored in a Problem-Based Learning (PBL) framework, the event combined ethnographic research, iterative prototyping, and intercultural collaboration. The paper critically examines the hackathon format as a pedagogical tool, exploring both its potential and its limitations. Particular attention is given to the dynamics of international teamwork, the challenges of inclusive design, and the tension between creativity and feasibility. It is argued that such educational interventions not only foster technical skills, but also cultivate empathy, cross-cultural understanding, and deeper engagement with complex real-world issues.

CCS Concepts

• **Human-centered computing** → **User studies**; *HCI design and evaluation methods*; • **Computing methodologies** → *Virtual reality*.

Keywords

Hackathon; PBL, robots; virtual reality; education; training

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1 Introduction

Hackathons have become increasingly recognized in higher education as effective environments for cultivating creativity, teamwork, and rapid problem-solving [Kamariotou and Kitsios 2022]. Originally developed in the tech industry to generate fast, functional prototypes, the format has expanded beyond coding competitions to address complex challenges in healthcare [Poncette et al. 2020], social innovation [Toros et al. 2022], and education [AlQallaf et al. 2024]. Their time-constrained, collaborative nature pushes participants to think creatively, iterate quickly, and work across disciplines—all essential skills for future engineers [Medina Angarita and Nolte 2020].

In parallel, Problem-Based Learning (PBL) has gained prominence as a pedagogical approach that shifts the focus from passive reception of information to active engagement with real-world problems [Wood 2003]. PBL encourages students to explore open-ended questions in teams, conduct independent research, and develop solutions rooted in context and user needs [Wells et al. 2009]. It supports not only the acquisition of technical knowledge but also the development of soft skills such as communication, adaptability, and critical thinking [Strobel and Van Barneveld 2009].

Combining these two models—hackathons and PBL—results in a hybrid format often referred to as a PBL Hackathon [Kumalakov et al. 2018]. In this structure, the fast-paced, outcome-oriented culture of the hackathon is paired with the reflective, user-centered mindset of PBL. Participants are encouraged not just to build, but to understand: to explore the underlying causes of a problem, identify the needs of stakeholders, and evaluate the broader impact of their solutions. Structured mentoring, user research, and iterative feedback loops are often built into the experience, ensuring that both learning outcomes and innovative results are achieved [Horton et al. 2018].

One of the most valuable features of the PBL Hackathon format is its capacity to foster interdisciplinary and intercultural collaboration. Unlike traditional classroom settings, which are often siloed by discipline or nationality, these events bring together diverse participants with varying backgrounds, skills, and perspectives. This mirrors professional engineering contexts, where solving global challenges demands not only technical acumen but also empathy,

cultural sensitivity, and the ability to communicate across boundaries.

This paper presents a case study of such an event: an international PBL Hackathon co-organized by two technical universities in Europe and Asia. The challenge posed to students was socially grounded and technologically demanding: how might immersive virtual reality and mobile robotics be combined to promote physical activity among children on the autism spectrum? Working in interdisciplinary teams, students followed a full PBL cycle—starting with field observations in a special needs school, followed by ideation, rapid prototyping, and public presentation of their final solutions.

In addition to addressing a meaningful societal problem, the hackathon served as a testbed for applying immersive technologies in education. Participants worked with VR goggles, mobile robot chassis, microcontrollers, and sensors to develop interactive experiences. They had access to expert mentors, rapid prototyping infrastructure, and cross-cultural collaboration workshops—all embedded within the structure of the event. By analyzing this educational intervention from both a technological and pedagogical standpoint, this article aims to demonstrate how immersive technologies like VR and robotics can be integrated into engineering education in a way that promotes real-world impact, learner autonomy, and cross-disciplinary innovation.

2 Event Overview

2.1 Objectives of the Event

To introduce the challenge, participants were shown a short presentation illustrating the barriers that children with autism spectrum disorder (ASD) face in engaging with physical activity. It featured a quote from a 12-year-old boy with ASD: “I love football, but playing... I don’t think I can imagine a worse place for me to be!” This set the tone by underscoring the dual challenge: navigating both social and motor difficulties. The presentation also referenced research indicating that regular exercise supports individuals with ASD by enhancing coordination, attention, cognitive performance, and emotional regulation, while lowering anxiety and [Rafiei et al. 2021; Yarimkaya and Esenturk 2020].

Participants were then tasked with designing a solution that merged virtual reality and robotics—complementary technologies known to foster motivation, engagement, and interaction. The aim was to promote physical activity among students with special needs through inclusive, technology-supported experiences that are both accessible and enjoyable. They were asked to “connect these two worlds” for learners with developmental differences, encouraging playful, movement-based interaction through creative and empathic design.

2.2 Participants

A total of 42 engineering students participated in the event: 17 from a technical university in Asia and 25 from a European counterpart. Students were recruited through internal calls at both institutions. Among the selected participants were 5 women and 37 men, aged between 20 and 30 years (with a mean age of 22 and a standard deviation of 2). The European students represented a range of technical disciplines: 2 from biomedical engineering and technologies, 7 from electronics and telecommunication engineering, 6 from automation

and robotics, and 10 from computer science. The Asian students were primarily enrolled in mechanical and robotics engineering programs. To encourage interdisciplinary and intercultural collaboration, the organizers formed six balanced teams, each consisting of both Asian and European students with complementary areas of expertise.

2.3 Organizational Infrastructure and Mentorship

The hackathon took place at a dedicated innovation facility, which offered participants access to a full suite of modern tools and development spaces. The challenge was supported by technologies that enabled the seamless integration of immersive virtual environments with physical robotics.

Each team had access to head-mounted displays, including Meta Quest 3 and Meta Quest Pro, for creating and testing VR scenarios. For robotics and sensing, students used Arduino and Raspberry Pi microcontrollers, motion and distance sensors, cameras, and touch-based input systems such as Makey Makey. They also had access to motorized robot chassis with omnidirectional wheels, allowing for the creation of mobile, interactive prototypes.

To support prototyping, the lab provided 3D printers with a variety of filament types, laser cutters, CNC milling machines, and a thermoforming machine. A dedicated electronics workshop enabled teams to assemble and test custom circuits, combining mechanical and digital components into cohesive systems.

The IT infrastructure included high-performance computers optimized for VR development and a stable network for real-time testing and device communication. Throughout the event, teams received ongoing technical support from staff members, who also ran short training sessions to onboard participants to the equipment.

Throughout the hackathon, participants had access to academic mentors, technical experts, and design facilitators who supported the development process. Guidance was available in the areas of virtual reality, robotics, inclusive design, and user experience. Although students were allowed to remain in the facility around the clock to work on their projects, mentor support was formally available for up to four hours per day. Participants also had direct access to a special education school, including its staff—teachers and psychologists experienced in working with children on the autism spectrum—as well as to parents and selected children with ASD. These contacts provided domain-specific input and enabled teams to gather practical information relevant to the challenge.

2.4 Event Logistics

The Hackathon was organized as an intensive, 12-day program combining remote and on-site activities (see Fig. 1). It began with an initial meeting during which participants were introduced to the challenge topic, organizational and safety rules, available equipment and infrastructure, as well as the scope of mentor support. The day also included ice-breaking and brainstorming sessions to establish intercultural team cohesion.

Participants then engaged in personal research, followed by an ethnographic field visit to a special education primary school. During this visit, they met with teachers, psychologists, parents, and children with ASD. The visit provided a first-hand opportunity to

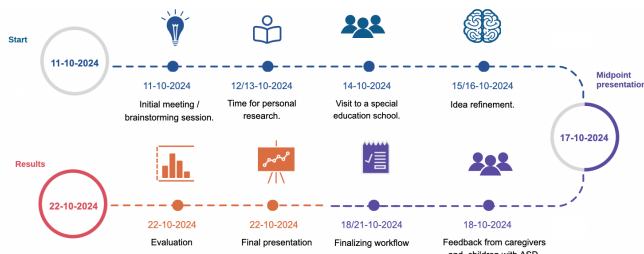


Figure 1: Timeline of the hackathon showing the key stages of the event.

learn about the daily challenges and needs of neurodiverse learners. Participants listened to the perspectives of educators and caregivers, and asked questions to understand the sensory, cognitive, and motivational barriers these children face in relation to physical activity. Based on these insights, teams refined their project ideas and selected final concepts. Working within their interdisciplinary groups, participants applied familiar PBL tools to structure their ideas and define clear development goals.

Midway through the program, participants received formative feedback from educators, parents, and children with autism, who reviewed early prototypes and shared their impressions. This input proved essential in identifying both strengths and shortcomings in the proposed solutions. Teams then had several days to revise and improve their concepts based on this feedback, finalize implementation workflows, and polish their prototypes.

The event culminated with final presentations and evaluations. Teams showcased their completed projects, explained the rationale behind their design choices, and participated in moderated discussion sessions focused on lessons learned, interdisciplinary collaboration, and future development potential.

2.5 Evaluation Methodology

The evaluation of teams during the hackathon followed a structured framework, in line with recommendations from the literature on collaborative assessment and participatory design [Brown et al. 2009; Lerchenfeldt and Taylor 2020], and was designed to provide a comprehensive and impartial judgment of the proposed solutions. The solutions were evaluated according to ten criteria: (1) general solution concept, (2) quality of the mechanical part, (3) quality in the field of robotics, (4) quality of IT solutions, (5) use of sensors, (6) use of behaviors and functions, (7) quality of the robot solution, (8) quality of the virtual reality solution, (9) user engagement and interaction, and (10) adaptability and personalization.

Each team member evaluated all other teams' projects using the established criteria. To encourage reflective thinking, participants were instructed to compare each solution with their own team's work, thereby engaging in a form of indirect self-assessment. The peer evaluation scores were then averaged across all student assessors to produce a single score per criterion for each team. In parallel, six domain experts independently assessed all projects using the same evaluation framework. Each evaluator group—each expert, the therapist, and the parents—was given the same weight as one team's aggregated peer assessment, ensuring a balanced

integration of professional, therapeutic, parental, and peer perspectives. To incorporate the end-user perspective, parents of children with autism—invited as external stakeholders—also participated in the evaluation. Their feedback was aggregated into a single stakeholder score, representing the lived experience of families directly affected by the proposed solutions. Finally, representatives from the teaching community contributed their assessments, with particular emphasis on the pedagogical relevance and educational value of each project. These educator evaluations were likewise included in the final scoring process, ensuring that the solutions aligned with broader learning and inclusion goals. To support the evaluation process, a web-based tool was developed for real-time scoring and result aggregation, ensuring transparency and enabling timely decisions by the organizing committee.

2.6 Post-Event Evaluation Questionnaire

To gain deeper insight into participants' experiences and learning outcomes, a post-event questionnaire was administered at the conclusion of the hackathon. It included both open- and closed-ended questions covering key areas such as personal goals and skills development, teamwork and intercultural collaboration, division of tasks and communication tools, use of equipment, perceived challenges, and educational impact. Participants were also asked about their interactions with mentors and stakeholders, the influence of the visit to a special needs school, and the relevance and potential applicability of their final projects. Additional questions addressed time management, overall organization, and willingness to engage in similar future initiatives. Responses were collected anonymously via an online survey tool and served both formative and summative functions—informing future improvements while providing insight into the event's educational effectiveness. A full list of questionnaire items is provided in Appendix A.

3 Results

3.1 Prototypes

3.1.1 Prototype 1 (P1): The VR environment was designed to deliver a first-person perspective from the robot's point of view. Users wearing the VR headset could see a real-time video stream transmitted from the robot's onboard camera, effectively immersing them in the robot's physical surroundings. The robot featured a dragon-shaped 3D-printed body (see Fig. 2 A). It was equipped with a front-facing camera and a mobile chassis, allowing it to move around in real-world space. The core interaction relied on immersive teleportation. The user could control the robot remotely by moving their body, using motion-based input to steer it while observing its live camera feed through the VR headset.

3.1.2 Prototype 2 (P2): The VR environment is set in an amusement park, where a virtual cat encourages the user to perform physical exercises through engaging, movement-based interactions. The physical robot prototype was a compact, wheeled device designed to resemble a stylized cartoon cat (see Fig. 2 C). Its front panel featured a cat face composed of large, illuminated eyes and a smiling mouth. The robot was designed to move left and right along a wooden rail, traveling from one end to the other. This motion directly mirrored the movements performed in the VR experience,

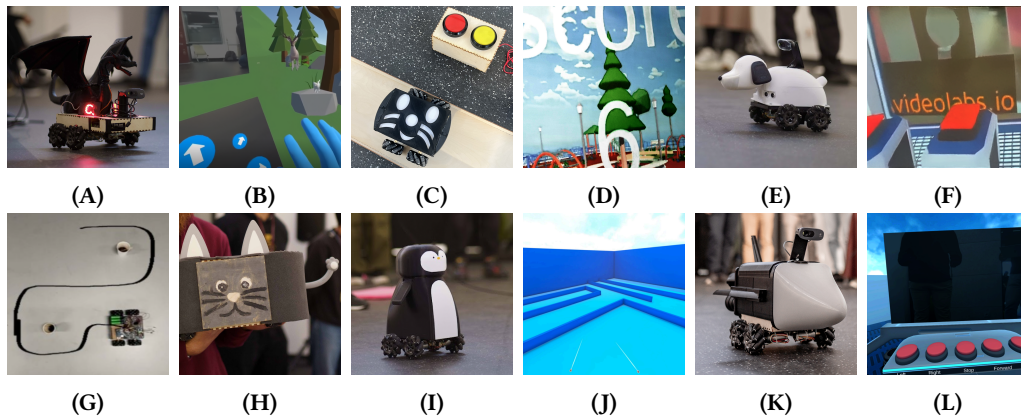


Figure 2: A set of images showing six prototypes, each with a robot (left) and its corresponding VR application (right): (A–B) Prototype 1, (C–D) Prototype 2, (E–F) Prototype 3, (G–H) Prototype 4, (I–J) Prototype 5, and (K–L) Prototype 6.

reinforcing the connection between virtual actions and real-world feedback. Additionally, the robot could be controlled using external buttons, prompting the VR user to replicate those same directional movements. This bidirectional control enabled a cooperative mode of play, allowing two users—one in VR and one outside—to interact and move in sync.

3.1.3 Prototype 3 (P3): The VR component offers a first-person perspective from the robot's onboard camera. The immersive view is streamed directly to the user's headset, allowing them to experience the environment from the robot's point of view. The robot has a compact, rounded form inspired by the shape of a dog (see Fig. 2 F). Its soft, approachable design includes a comfortable texture and no sharp edges. Key hardware components include ultrasonic sensors for obstacle detection, a front-facing camera for video streaming, and a battery module safely hidden. The robot streams live video to the VR headset, allowing users to navigate it remotely. Movement commands issued in the VR environment directly control the robot's actions in real space.

3.1.4 Prototype 4 (P4): This prototype did not include a traditional immersive VR environment. Instead, it focused on a mobile application that allowed children to draw a path on a touchscreen (tablet or phone). The interface was designed to be simple and intuitive, encouraging engagement through visual interaction and active participation in task design. The app incorporated real-time camera input and dynamic scaling based on the robot's physical surface dimensions. The robot had a friendly and customizable appearance, with a soft protective outer layer made of sponge material for safety (see Fig. 2 H). The base could be adapted to different animal-like covers depending on children's preferences. The key feature of this system was the ability for users to draw a custom path on the mobile app, which was then translated into movement commands for the robot.

3.1.5 Prototype 5 (P5): The team designed a virtual maze environment where children are tasked with navigating an obstacle course by guiding a robot. The interface includes both VR and AR modes, allowing users to experience the scenario either through immersive

headsets or augmented reality overlays on tablets or smartphones. In the VR/AR environment, users are required to plan the robot's path through the maze by interacting with virtual elements. The physical robot resembles a cheerful penguin (see Fig. 2 I), with a rounded, child-friendly form and movable wings animated via a slider crank mechanism. Its body flaps its wings when tasks are successfully completed. Children use the VR or AR application to visualize and interact with the maze. Their actions translate into commands that control the real-world robot via WiFi. The robot acts out movements and gestures based on progress in the game, serving both as a feedback device and as a physical embodiment of the digital experience.

3.1.6 Prototype 6 (P6): The application places users in a stylized outer-space setting guided by a friendly virtual cat-like astronaut. Within this immersive VR environment, children are invited to complete space-themed tasks such as navigating through planetary surfaces and collecting virtual objects. These activities are designed to encourage full-body motion and improve motor engagement. The physical component is a mobile robot resembling a small rocket-shaped vehicle (see Fig. 2 K). The robot includes movement capabilities and visual signals to provide feedback and enhance engagement. Actions performed in VR trigger corresponding robot movements and behaviors. A custom-built VR control panel enables the user to manage robot actions during gameplay, creating a loop of continuous interaction between physical and virtual elements.

3.2 Hackathon Outcomes

The results reflect multi-source evaluations based on shared criteria. Table 1 shows the aggregated scores.

The table reveals substantial variation in how teams assessed each other, with some assigning disproportionately high or low scores—possibly reflecting strategic bias. In contrast, evaluations from external reviewers, including experts (E1–E4), parents (P), and teachers (T), followed a more consistent pattern. Team 2 stood out as the clear leader, receiving the highest overall score and consistently strong evaluations across all external groups. Teams 1 and 3 were

Table 1: Hackathon Team Evaluation Results. Columns P1–P6 are teams; E1–E4 expert scores (E1–E2: VR experts, E3–E4: robotics), P parents, T teachers, and Avg the final average. For expert scores, 0 denotes the minimum acceptable level; values < 0 mean the project does not meet expert expectations.

	P1	P2	P3	P4	P5	P6	E1	E2	E3	E4	P	T	Avg
P1	54.6	-32.4	12.8	75.1	8.5	52.2	68.3	22.5	54.7	70.2	83.5	74.9	45.4
P2	58.6	9.2	-39.2	70.8	3.9	78.8	76.4	67.2	51.3	37.1	50.1	34.8	41.6
P3	10.9	9.5	3.4	65.4	-10.5	40.4	40.4	48.8	65.6	49.7	99.4	50.1	39.4
P4	-32.1	10.0	9.4	54.0	4.0	30.4	47.5	31.3	62.0	7.4	-6.7	46.1	21.9
P5	54.5	37.1	53.1	54.0	-7.1	-42.4	28.4	21.9	9.4	9.5	0.8	28.1	20.6
P6	31.0	20.5	55.0	-72.1	2.8	32.8	47.2	-18.7	28.7	31.4	21.5	54.5	19.6

also rated favorably, though peer assessments somewhat reduced their final averages. In contrast, Teams 4 and 6 received low ratings from both experts and stakeholders, indicating limited alignment with user needs. The inclusion of diverse evaluator perspectives helped mitigate internal bias and yielded a more robust project ranking.

3.3 Participant Feedback and Reflections

3.3.1 Personal Goals and Skills. A comparison of participants' initial goals and their self-reported skill development reveals a general alignment between expectations and outcomes (see Table 2). The most frequently stated goal was to improve communication skills—especially in English—while the most commonly developed skills were related to programming, Unity, and teamwork. Notably, categories such as learning VR/Unity and developing teamwork skills appear in both columns, suggesting that participants not only prioritized these areas but also made progress in them. The diversity of skills gained—from electronics to self-awareness—illustrates the hackathon's potential to support both technical and interpersonal learning. Overall, the table reflects the multifaceted nature of the experience and its effectiveness in addressing a broad range of learner objectives.

3.3.2 Teamwork Structure. As shown in Figure 3, participants generally evaluated teamwork positively, with most rating it as good or excellent. Despite the international composition of the teams, group cohesion was not perceived as a major challenge. Figure 4 indicates that most respondents found the process of team integration easy or moderate. Only a minority described it as difficult, suggesting that cultural differences had limited negative impact on collaboration.

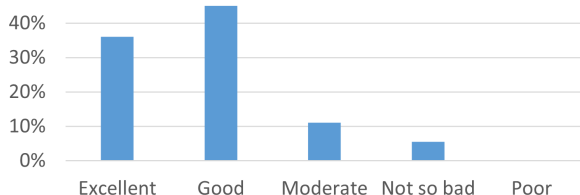


Figure 3: How would you rate the teamwork during the hackathon?

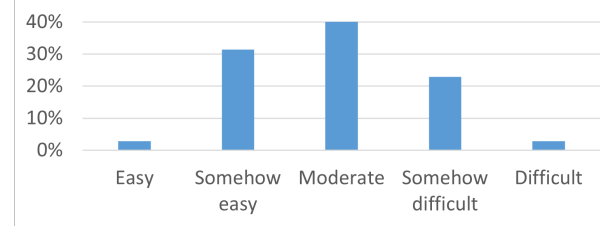


Figure 4: Given the different cultural backgrounds of the participants, did you find it challenging to form a cohesive team, or did it come easily?

Table 3 summarizes key aspects of teamwork reported by participants. Teams typically consisted of 6–7 members, which was considered an effective size for collaboration. Random assignment to international groups was generally viewed positively, as it enabled participants to connect across cultural and academic boundaries. Respondents highlighted the value of meeting new people, encountering diverse perspectives, and experiencing different styles of communication. Several noted that randomization prevented homogeneous group formation and fostered more inclusive teamwork.

Most participants reported no significant challenges in navigating cultural differences. Many did not perceive these differences as barriers or considered them negligible within the context of technical collaboration. Several attributed this to the shared project focus, which helped bridge gaps in communication styles and expectations. In cases where cultural or language-related difficulties arose—most often linked to English fluency or conversational dynamics—participants applied pragmatic strategies such as using translation tools, speaking clearly, or asking clarifying questions. A few emphasized the importance of empathy and openness, noting that they consciously adapted to diverse speaking styles or addressed cultural differences directly to prevent misunderstandings. Participants also described their peers as "gentle," "friendly," and "respectful," and many viewed collaboration itself as a valuable form of cultural exchange. Informal conversations and small talk were frequently mentioned as important in building mutual understanding.

Participants reported that working in unfamiliar teams posed initial challenges—such as language barriers, differing expectations, or varying levels of prior knowledge—but these were often reframed as learning opportunities. For some, the experience encouraged

Table 2: Comparison of participants' initial goals and skills developed during the hackathon

Goal Category	Mentions (Goal)	Skill Category	Mentions (Skill)
Improve communication skills (esp. in English)	17	Learn VR / Unity	9
Make international friends	6	Learn robotics / electronics	8
Overcome language or cultural barriers	4	Develop teamwork skills	7
Complete a project from start to finish	5	Programming / application development	10
Learn VR / Unity	9	3D modeling	4
Learn robotics / electronics	8	Electronics / microcontrollers	4
Create a mobile or VR app	3	Project management / task delegation	2
Develop teamwork skills	7	Self-awareness / breaking barriers	2

Table 3: Frequency of Reported Teamwork Aspects

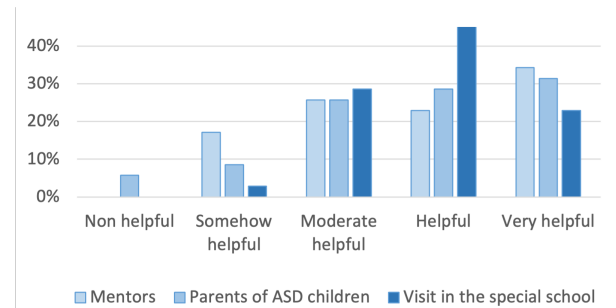
Aspect	Number of mentions
Cultural differences	~30 positive/neutral; ~8 mention issues
Random team assignment	~35 positive; ~6 negative
Team size	~30 said 'appropriate'; ~5 said 'too large'
Sub-team structure	> 40 teams reported subgroups
Team vs. individual tasks	All teams mentioned some task division
Communication tools	Discord (~30), LINE (~10), others (~5)

stepping out of their comfort zone, improving interpersonal communication, and developing adaptive problem-solving skills. A few participants noted that mismatches in expertise or communication styles occasionally hindered productivity, and that outcomes might have been more refined had they chosen familiar collaborators. However, such views were in the minority.

Sub-teams were a dominant organizational strategy. Most groups divided into smaller units focusing on distinct tasks such as robot construction, VR/game development, hardware integration, or 3D modeling. Conceptual work and decision-making were usually carried out as a full team, while technical tasks were delegated individually or in pairs based on experience and interests.

Discord was the most widely used tool for intra-team communication, followed by LINE. Other tools included Miro, Messenger, and online translators. This mix of informal and digital tools supported ongoing knowledge exchange, task coordination, and mutual assistance.

3.3.3 Stakeholder Interaction. As shown in Figure 5, most participants rated the visit to the special school and the presentation session with parents as helpful or very helpful in shaping their project ideas. The support from mentors was also positively evaluated, though with a slightly broader range of responses—some students found it moderately useful, while others highlighted its critical role in clarifying technical or design-related questions. The school visit and parent session provided essential context on the needs, emotional responses, and daily experiences of children with ASD. Key insights included the recognition that each child is different, that emotional age may differ from chronological age, and that technology-based interactions must be tailored to sensory sensitivities and cognitive preferences. Many participants noted that the experience deepened their empathy and helped refine their designs—for example, by simplifying robot control methods, adjusting game mechanics, or rethinking visual and thematic elements to

**Figure 5: Did the input from mentors, parents of children with ASD, and the school visit help you develop your project concept?**

better match children's comfort levels. For some, it was their first direct engagement with neurodiverse children, and they described the encounter as transformative. A small number of respondents also observed inconsistencies between feedback from parents of children with ASD and the children's teachers, underscoring the value of engaging both groups—each with direct, day-to-day contact with ASD—throughout the design process to align goals. Overall, while mentor guidance supported technical implementation, stakeholder engagement helped ground the projects in real user needs—highlighting the importance of integrating both pedagogical input from teachers and experiential input from families in future iterations.

3.3.4 Equipment and Resources. Participants reported varying levels of engagement with the available fabrication tools and resources (see Figure 6). The most intensively used items included the VR set, electronics workshop, and hand tools, reflecting the dual focus on virtual and physical prototyping. Devices such as 3D printers, laser cutters, and graphical resources were also frequently used, although less consistently. In contrast, tools like the milling machine, Makey Makey, and thermo-forming equipment saw minimal to no use in most teams. These patterns indicate a strong emphasis on accessible, versatile tools that align with rapid prototyping workflows.

3.3.5 Time allocation. Participants were asked to estimate their average daily time commitment to the hackathon and how it was divided between team discussions and individual work. As shown in Figure 7, most reported spending 6–8 hours or more per day

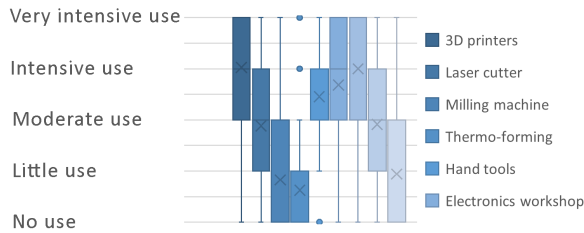


Figure 6: Reported use of prototyping tools during the hackathon.

on the project. Among these, the majority devoted 3–5 hours to team discussions, indicating a strong emphasis on coordination and planning. While group communication played a central role, many participants also reported substantial time spent on individual tasks such as coding, modeling, or electronics. This points to a hybrid working model, where collaborative design was followed by self-directed technical implementation. A few respondents indicated minimal discussion time (0–2 hours) despite high overall commitment, suggesting that in some teams, work may have been carried out in parallel rather than through continuous interaction.

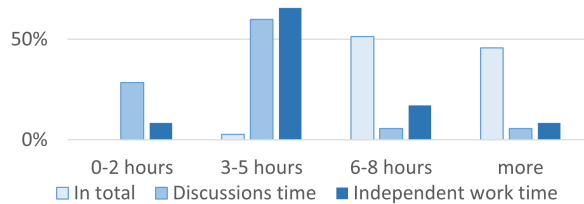


Figure 7: Self-reported time spent on project work: total, discussion, and independent effort

3.3.6 Motivation and Learning. As shown in Figure 8, most participants reported that the hackathon positively influenced their motivation to learn and explore new topics. This aligns with qualitative responses on how specific elements of the event supported learning and skill development. Participants most frequently pointed to

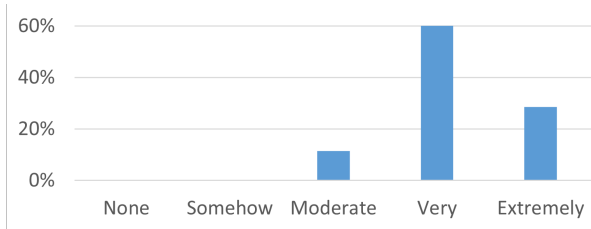


Figure 8: How participating in the hackathon increase your motivation to learn and explore new topics?

brainstorming as the primary problem-based learning (PBL) method that helped them initiate and shape their projects. Several groups described starting with open discussion sessions to collaboratively

generate ideas, identify promising directions, and divide tasks. This early ideation stage helped clarify roles and encouraged team members to focus on specific aspects of the solution.

The interdisciplinary team structure was also identified as a key factor. Many participants noted that working with peers from different academic or cultural backgrounds enriched their approach, introduced them to new ways of thinking, and fostered mutual learning.

In terms of technological discovery, 29 respondents indicated that they were exposed to new tools or platforms. Frequently mentioned examples included Unity, API integration, GitHub, and 3D CAD software. A few participants reported using VR creation tools for the first time and expressed interest in applying them in future projects at their home institutions. Some encountered unfamiliar equipment or software—such as 3D printers or HUD (head-up display) technologies—and described this exposure as both motivating and inspiring.

While 3 respondents stated they did not learn any completely new technologies, the majority emphasized that most aspects of implementation—from hardware integration and programming to VR development—were novel experiences. These tools and methods were considered relevant not only to the hackathon but also to their broader academic and professional growth.

3.3.7 Broader Impact. Participants offered mixed opinions on whether their final projects could effectively encourage autistic children to participate in sports (see Figure 9). Some expressed confidence that their prototypes had the potential to promote physical activity in an enjoyable and accessible way. Others emphasized the simplicity of interaction, emotional engagement through play, and adaptability to different user needs as key strengths. Several respondents noted that while the core ideas were promising, their prototypes remained incomplete and would require further refinement, additional features, or better coordination between VR and physical systems to be truly effective. A few participants believed their solutions were unlikely to encourage sports participation, either due to limited physical movement in the game or a lack of alignment with sports-related goals. The hackathon had a strong and gen-

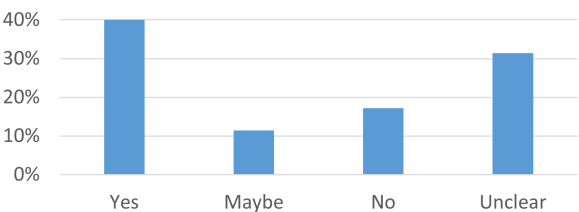


Figure 9: Perceived impact on promoting physical activity in ASD.

erally positive impact on participants' views of collaboration and networking. 31 respondents reported meeting new people, forming friendships, and gaining valuable interdisciplinary contacts. For 21 participants, it was their first time collaborating across language or cultural barriers, which broadened their perspective and increased their appreciation for teamwork. 14 others highlighted how the

shared experience of overcoming technical and communication challenges fostered strong interpersonal bonds and inspired professional growth. Only 5 participants reported no major networking outcomes, although even among these, collaboration was still regarded as a meaningful learning experience.

3.3.8 Event Organization. Participants overwhelmingly rated the overall organization of the hackathon positively, with most responses ranging from "very good" to "excellent" (see Figure 10). The atmosphere of collaboration, the opportunity to work across cultures and disciplines, and the hands-on nature of the activities were frequently cited as highlights of the experience.

The most memorable aspects mentioned by respondents included team bonding moments, the first successful test of a working prototype, and social integration events such as the welcome party or shared meals. Several participants emphasized that the experience of getting their robot or VR application to finally work—often after prolonged effort—was particularly rewarding. Suggestions for improvement focused primarily on technical logistics (e.g., better internet access, clearer instructions, more tools and equipment) and project clarity (e.g., better initial definition of goals or more realistic scope). A few participants also noted issues with scheduling or difficulty in aligning feedback from different stakeholder groups (e.g., teachers vs. parents).

When asked whether elements of the hackathon could be integrated into academic education, 34 participants expressed strong interest in including similar project-based and interdisciplinary activities in their regular courses. They highlighted the motivational value of working on real-world problems, the benefits of team-based learning, and the potential of VR technologies in education. 19 respondents explicitly referred to the hackathon as a more engaging and effective form of PBL than what they had previously encountered in academic settings. Finally, 39 participants declared their willingness to participate in similar events in the future, confirming the educational and social value of the hackathon format.

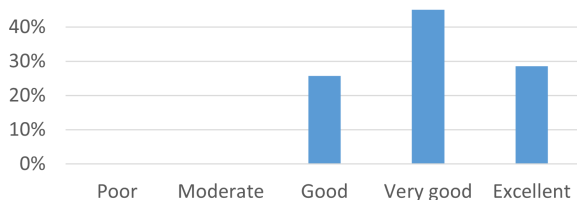


Figure 10: How would you rate the overall organization of the hackathon?

4 Conclusion

Despite the diversity of backgrounds and ideas, several project outcomes converged on similar design patterns—particularly scenarios involving mazes, obstacle navigation, and mobile robots equipped with cameras. While this recurring solution motif demonstrates a shared intuitive understanding of the challenge, it may also suggest that participants became fixated on familiar tropes rather than

fully exploring novel or user-driven concepts. This tendency could reflect both the time constraints of the hackathon format and the participants' limited prior exposure to inclusive design methodologies.

Beyond the repeated use of maze-like scenarios, several deeper patterns emerged. Most teams treated the robot primarily as a navigational object rather than a social actor, missing opportunities to explore human-robot interaction as a form of engagement—particularly relevant when designing for autistic children. Additionally, although a range of sensors and actuators were available, few teams incorporated meaningful sensory feedback tailored to users' needs. Similarly, many VR scenarios replicated physical spaces instead of reimagining environments in ways that could enhance motivation or sensory comfort. These choices suggest a tendency to use immersive technologies conservatively, missing opportunities to leverage their full transformative potential. In some cases, the level of physical exertion required in the activities was minimal, potentially weakening the effectiveness of the solutions in promoting actual movement and exercise.

Another notable observation concerns the evaluation process. While the multi-source framework added rigor, the relatively complex scoring system—featuring ten criteria across multiple evaluator groups—may not have been fully understood by all participants. As a result, some inconsistencies in peer scoring emerged, potentially skewing outcomes. Future iterations might benefit from simplified evaluation rubrics or more structured calibration sessions to ensure shared understanding.

Interestingly, cultural differences did not emerge as a major barrier. Participants largely reported smooth collaboration regardless of nationality. However, few respondents cited intercultural exchange as a central aspect of their experience, suggesting that while cultural diversity was present, it was not always actively engaged as a learning resource. This presents an opportunity for future programs to more intentionally leverage cross-cultural dialogue as part of the educational design.

Participants also expressed mixed confidence in the impact of their final prototypes. While most saw value in their core ideas, many emphasized that the implementations remained incomplete and would require significant refinement to be viable in real-world educational or therapeutic contexts. This pragmatic self-assessment reflects a mature understanding of the prototyping process, but also reveals a gap between the participants' ambitions and the limited fidelity achievable within the hackathon timeframe.

While the event was grounded in a Problem-Based Learning (PBL) framework, the actual use of structured PBL methods during project development was limited. Brainstorming was the only PBL tool frequently mentioned in participant responses, with little reference to methods such as stakeholder mapping, problem reframing, or iterative hypothesis testing—despite students having prior exposure to these techniques in coursework. This raises the question of whether participants perceived such methods as unnecessary under time pressure, or lacked confidence in applying them outside a classroom setting. Either interpretation points to a relevant pedagogical insight: incorporating short, facilitated PBL exercises early in the hackathon might increase the likelihood that teams internalize and apply these strategies throughout the project.

Finally, there was a clear alignment between the goals participants set at the outset and the skills they reported developing. Communication, teamwork, and technical abilities—such as Unity and electronics—featured prominently in both categories. This indicates that the hackathon served not only as a technical challenge, but also as a meaningful learning experience. Still, many participants expressed a desire for more time, clearer scopes, and opportunities for longer-term development, suggesting that the event sparked educational ambitions beyond what its short format could accommodate.

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A Post-Event Questionnaire: Full List of Questions

Table 4: Post-event evaluation questionnaire used to assess participants' experiences, learning outcomes, and team dynamics during the hackathon.

Category	Question(s)
Personal Goals and Skills	What goals did you set for yourself at the beginning of the hackathon? Were you able to achieve them?
Teamwork	What skills did you develop or enhance during the event? Which of these are the most important to you? How would you rate the teamwork during the hackathon? Given the different cultural backgrounds of the participants, did you find it challenging to form a cohesive team, or did it come easily? How did you navigate any cultural differences during the hackathon? Do you think being assigned to mixed groups from the start, rather than choosing your own team, had a positive or negative impact on your experience? Why? Was the number of people in your group appropriate for effective collaboration, or did you feel it was too large or too small? Please explain. What tasks did you complete as a team and which ones did you work on independently? Were there sub-teams for specific tasks? How was this division organized? Did you use any specific tools for communication within your group? If so, which ones were most effective?
Stakeholder Interaction	How did the visit to the special school and conversations with teachers of autistic children influence your project? Did you gather the necessary information during the school visit and parent presentations? What insights were most valuable?
Equipment and Resources	How did mentors and parents of ASD children influence your work? Which of the following tools did you use during the hackathon: 3D printers, laser cutter, milling machine, thermo-forming, hand tools, electronics workshop, VR set, graphical resources, Makey Makey?
Time Allocation	On average, how many hours per day did you work on the hackathon? How much of that time was spent in team discussions?
Motivation and Learning	How much of that time was spent working independently? How did participating in the hackathon increase your motivation to learn and explore new topics? What was the biggest challenge you faced, and how did you overcome it? What was the most enjoyable part of the experience?
Project Impact	Which PBL tools or methods helped you most in completing your project? Did you discover any new technologies useful for the future? Do you think your final project could effectively encourage autistic children to participate in sports? Why or why not? What impact did the hackathon have on your perspective on collaboration and networking?
Event Organization	Did you establish any important new contacts? How would you rate the overall organization of the hackathon? What could be improved in future editions? What part of the hackathon do you find most memorable? What ideas do you have for applying this experience in education? Could any elements be integrated into regular academic courses? Would you be interested in participating in similar projects in the future?